FLEXIPLE PRODUCT INTERN ASSIGNMENT · PRE-ALPHA PROTOTYPE

# GCC Atlas — India’s GCC ecosystem, made searchable.

Provenance-first city, policy and company comparison for India capability-centre decisions: every number carries its source, its date, and its counting denominator.

WORKING TITLE — name collision flagged (S38)

COMPOSITES FROM EVIDENCE — NO INTERVIEWS CONDUCTED

| GCCS (PARENT-LEVEL) | GCC UNITS | OF 2025 OFFICE ABSORPTION | GCC REVENUE FY26 |
|---|---|---|---|
| 2,117 | 3,728 | 39% | $98.4B |
| FY2026 · S01/S02 | FY2026 · S01 | CBRE · S20 | Nasscom–Zinnov · S02 |

Prototype runs from repo: `npm run setup && npm run seed && npm run dev:api` + `npm run dev:web` · Walkthrough: deployment/demo-walkthrough.md

# The problem: fragmented intelligence, costly decisions

India’s GCC ecosystem is large, fast and definitionally messy — the decision layer above company directories is nowhere assembled.

GCCS PARENT-LEVEL FY2026

**2,117**

Nasscom–Zinnov · S01/S02

FY2026 REVENUE

**$98.4B**

same series · S02; 3 adds independent confirm

OF 2025 OFFICE-ABSORPTION IS GCCS

**39%**

CBRE leasing records · S20/S22

- **8+ state policies in 8+ formats.** Karnataka, TN, UP, Gujarat, Kerala… each publish subsidy tables in different shapes (payroll %, land subsidy, stamp duty, OPEX caps) — normalizing them is manual analyst work (S05–S19).
- **Counting regimes conflict.** 2,117 GCCs / $98.4B (Nasscom–Zinnov, S02) vs "$164B / 4.2M jobs" (Wizmatic via Statesman, S04 — low-confidence definitional outlier, registered as a conflict, never averaged).
- **The comparison is done by consultants on 3–6-week cycles.** A build-vs-lease decision uses brokerage PDFs + press releases + Excel (S01, S24); a defensible shortlist with citations takes weeks of analyst hours.
- **Source quality varies wildly across DIY databases.** DIY city directories contain row-level coding mistakes (GCC Village: ~473 Hyderabad vs ~100 Bengaluru rows, S35) that can severely misinform a location business case.

Everything above is cited; no market-size figure on this slide is our own estimate. All source IDs refer to research/source-register.csv.

STAKEHOLDERS (composite personas, evidence-based)

# Who feels the information asymmetry

PRIMARY — designed for

## Priya Raman

Location-strategy lead, MNC expansion desk

- Translates global headcount mandate → Indian city criteria
- Needs ranked candidate cities with talent, cost & subsidy facts
- Must defend shortlists to CFO, tax team & external consultants
- Fears "why does your slide contradict the broker report?" in review

Basis: 64% of GCC space take-up by her class of buyer (S20,S21); UK/US firms drive 37–45% of India office absorption (S24). Composites; no interviews conducted.

secondary — served

## Rakesh

State investment-promotion (IPA) officer

- Competes vs peer states with hard numbers, not policy slogans
- Wants a shareable, credible neutral link for prospective investors
- Fears third-party commercial tables painting the state unfairly

Basis: 8+ states run active GCC-acquisition playbooks with published targets (S05,S10,S13,S16). Secondary persona — distribution catalyst in v1.

secondary — served

## Maya

Provider / enabler BD lead (RE, legal, staffing)

- Wants verified, ready-to-engage corporate demand signals
- Currently pays $3.5K–$10K/year for commercial lead surfaces (S30)
- v1 gives her nothing to pay for — credibility & trust cultivated first

Basis: GCC Explorer monetizes exactly this class (S30); AIM watchlists purchased by this class (S32). Provider neutrality is an enforced rule.

Anti-personas by design: job-seekers (no JTBD) and competitor scrapers. Everything persona-labelled as [inference] through Phase-3 validation gates.

COMPETITIVE LANDSCAPE (verified, S30–S38)

# Current workarounds and where they stop

## GCC Explorer (AIM Research)

1,680+ GCC directory, daily analyst updates, contacts, watchlists. $3,500–$10,000 per seat cycle; free tier = 15 companies (S30/S31/S32).

**Their gap:** Not a city-DECISION tool: no policy-incentive normalization, no side-by-side compare with counting denominators (S30 pages reviewed).

## Consultancies (CBRE/JLL/Zinnov)

Authoritative bespoke decks; mandates; deep local nuance (S01, S20–S24).

**Their gap:** Time + high fees; 3–6 week project cycles; not self-serve at session speed.

## State portals + PIB PDFs

Primary-source truth; free access (S05–S19).

**Their gap:** 8+ formats; conflicting denominators; normalizing incentives is analyst hours.

## DIY directories (GCC Village et al.)

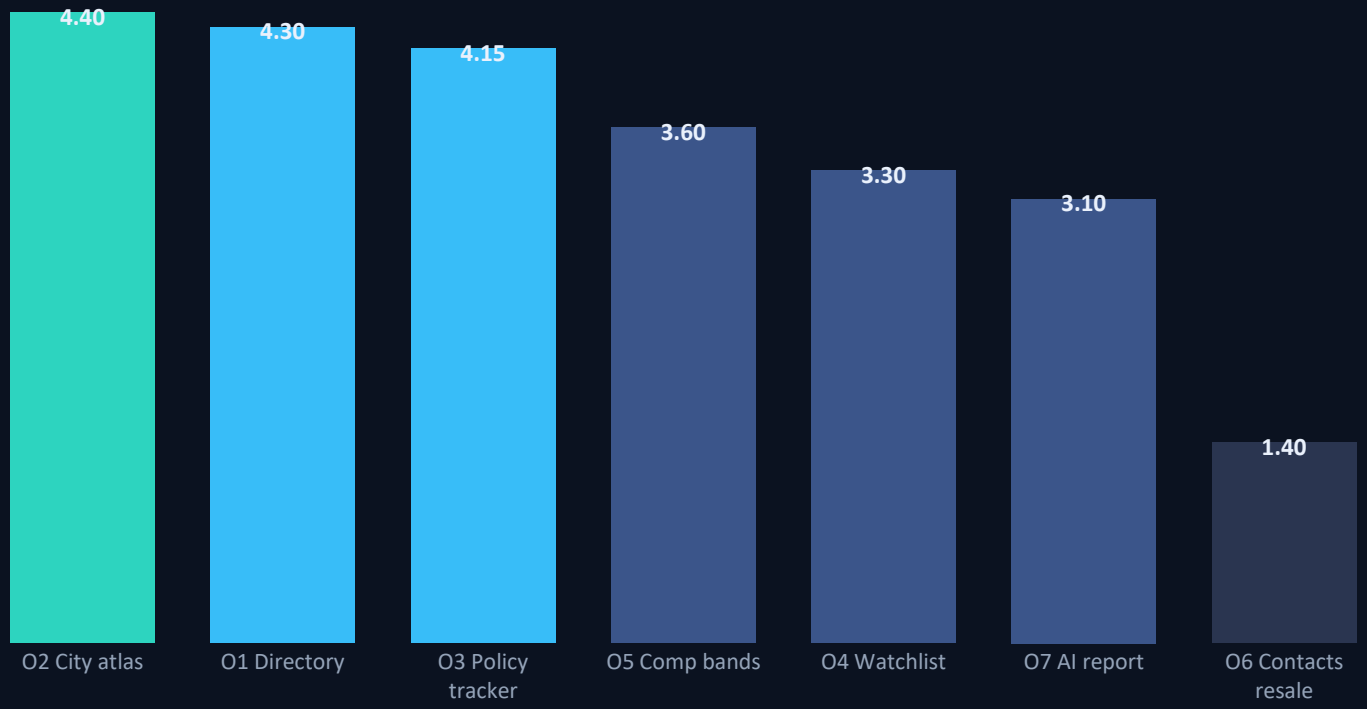
Free coverage slices; open directory tables (S35–S37).

**Their gap:** Row-level coding inconsistencies (S35); unverified data; zero provenance links.

**White space we occupy:** a free, provenance-linked city-vs-city comparison layer with explicit counting denominators — no verified analog exists across the scanned landscape.

SCORING (research/opportunity-prioritization.md)

# Opportunity map and prioritization




Scores = weighted TAM / underservedness / public-data feasibility / prototype fit / risk (weights 20/30/25/15/10). Method + rubric documented in research/opportunity-prioritization.md.

## WINNER: O2 — the decision layer

- **City atlas + location-comparison** is the most underserved, most buildable surface (O2 score 4.40).
- **O1 directory and O3 policy tracker ship as** supporting modules inside it.
- **Deliberately excluded:** contact-data resale (O6: name/legal risk, S38), watchlists/O4 (needs live crawler infrastructure), comp-band library (O5: licensing cost, Phase-3 candidate).
- **Falsification path:** if a real expansion desk says an ad-hoc spreadsheet already solves their need, we pivot to O4 (documented in research §6).

PRIMARY SEGMENT (evidence-based composite)

# Initial target customer and the job to be done

## Job statement (from primary persona)

> "Compare candidate cities for a new GCC — talent, policies, real-estate cost and sector peers — for a specific function, with every number carrying its counting denominator and its source."

- **JTBD 1 — Frame:** translate parent company mandate → objective city criteria (function, tier, cost ceiling).
- **JTBD 2 — Shortlist:** generate ranked candidate cities with verified facts (talent pool, Grade-A rents, subsidies).
- **JTBD 3 — Defend:** ensure shortlists survive CFO, tax desk, and consultant scrutiny with explicit citations.
- **JTBD 4 — Act:** identify exact IPA contacts and eligible subsidy bands (TN 30/20/10% payroll bands S06; UP ₹20k/fresher S10).

## Why this segment first

- **Documented demand density:** US/global firms drove 37–45% of recent office absorption; this class routinely pays for location advisory (CBRE/JLL advisory hours S20–S24; GCC Explorer $3.5K–$10K S30).
- **Severe workflow friction:** 8+ state policy formats + conflicting market counts = exactly the information asymmetry documented in market-research §2–3.
- **Reachable without paid ads:** via state IPA relationships + public expansion-team leadership desks [inference].
- **Measurable acceptance gate:** M1 time-to-answer metric (<10 min) is instrumentable end-to-end.

**[inference] items bound for Phase-3 validation interviews**

# From Fragmented Data to Defensible Decisions

STAGE 1

## 1. SCATTERED PUBLIC DATA

Unstructured Raw Input

- **Disparate PDF formats:** 8+ state Government Orders (G.O.s) with incompatible subsidy structures (S05–S19).
- **Brokerage PDF reports:** Fragmented real estate asking rents and vacancy rates across CBRE, JLL, Colliers (S20–S24).
- **Conflicting counts:** Nasscom 2,117 GCCs vs Wizmatic $164B/4.2M outlier (S02 vs S04).
- **Friction:** Analysts waste 3–6 weeks hunting, copying, and reconciling unformatted sources.

STAGE 2

## 2. NORMALIZATION & PROVENANCE

GCC Atlas Data Engine

- **Unit standardization:** Grade-A rents normalized to INR/sqft/mo; talent pools to headcount; subsidies to uniform schemas.
- **Definitional labeling:** Every metric explicitly tagged with counting denominator (e.g. parent entities vs units).
- **Source hashing:** Immutable link to publisher, publication date, verification date, and confidence level.
- **Seed linter:** Hard code invariant halts builds if any displayed figure lacks provenance (F6).

STAGE 3

## 3. EXPLORATION & CITATION

Interactive Decision Surface

- **Side-by-side matrix:** Compare 2–4 cities with explicit EST flags, metric definitions, and state policy clauses.
- **Bounded AI assistant:** RAG engine strictly constrained to retrieved evidence packs; hallucinated citation IDs stripped.
- **Recommendation split:** Factual evidence cleanly separated from strategic guidance; declines if data is thin.
- **Shareable sessions:** URL state (?ids=) and server-scoped persistence for cross-stakeholder collaboration.

STAGE 4

## 4. DEFENSIBLE DECISION

Intended Outcome [Hypothesis]

- **3–6 weeks → 1 session:** Target: condense initial location shortlisting from weeks to <10 minutes (M1 instrumented).
- **Board-ready audit trail:** Every cell, subsidy band, and quote carries clickable primary-source backing.
- **Zero broker bias:** Neutral comparator free of pay-to-play rankings or sponsored placements.
- **Credibility boundary:** [INTENDED OUTCOME — UNVALIDATED IN PHASE 2; scheduled for Phase-3 MNC desk trials].

Workflow maps research/market-research.md to product architecture. Stage 4 outcomes are explicitly tagged as intended targets awaiting Phase-3 validation.

WHAT IS BUILT (running pre-alpha prototype)

# The product and its differentiated value proposition

**One line:** evidence-first India GCC location intelligence — the decision layer above directories, below bespoke consulting ("free to look, defensible to reuse").

### City Intelligence

12 tracked cities × 6+ metrics; side-by-side compare of 2–4 cities w/ metric definitions, units and EST flags.

dataset: tracked records only (scope-labelled)

### Policy matrix

State incentives normalized vs source G.O.s; draft/final states distinct; "not yet tracked" ≠ 0 (Kerala dual targets co-labelled, S16 vs S18).

### Provenance everywhere

46 GCC records, 36 sources; every figure → source chip → publisher, URL, pub date, verification date, confidence.

seed linter fails build on sourceless values

### Cited AI assistant

Retrieval-pack answers; only retrieved evidence may be cited; hallucinated IDs stripped (tested); recommendations separated; declines when thin.

demo mode composes deterministically without key

### Saved research + share

Session-scoped server-side persistence (cross-session isolation tested); shareable deep links; CSV export.

### Event instrumentation

PRD §3.5 catalog (search, detail_view, compare_create, source_click, export, saved_create…) — hashed session keys; aggregate admin.

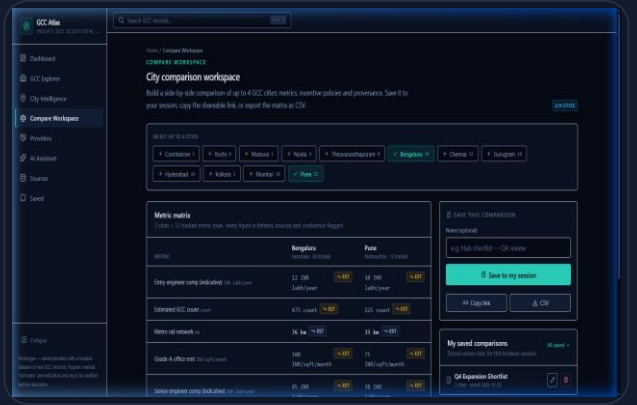
All coverage claims are dataset-scope-labelled; the app never claims the 2,117 national count as its own data.

SCREENSHOTS & MODULE RATIONALE (running pre-alpha build captures)

# Product walkthrough — genuine prototype captures

## 1. City Comparison Workspace

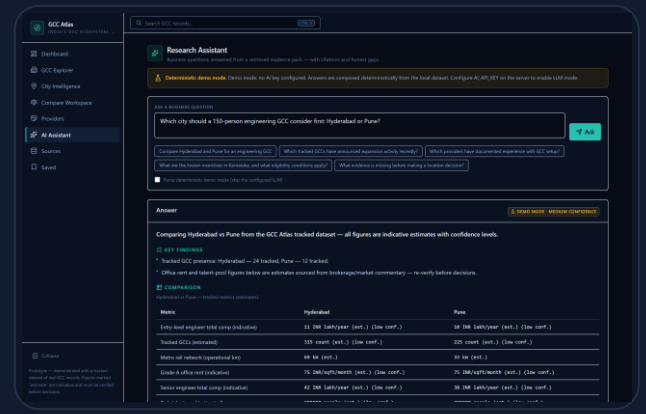
**DECISION MATRIX**




- **What it does:** Side-by-side metric matrix (talent pool, Grade-A rents, comp, infra) and normalized state incentive policies across 2–4 selected cities with shareable URL parameters and CSV export.
- **Problem:** Manual multi-week reconciliation across 8+ incompatible state government orders (G.O.s), brokerage PDFs, and ad-hoc analyst spreadsheets.
- **Why built this way:** Explicit EST flags, hoverable counting definitions, and transparent source citations ensure shortlists survive CFO and consultant scrutiny without hidden broker bias.

## 2. AI Research Assistant

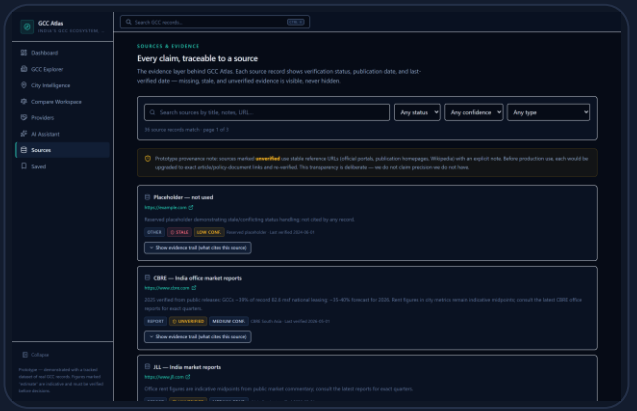
**BOUNDED RAG**




- **What it does:** Retrieval-augmented query engine answering multi-city and expansion questions using strict local evidence packs with verified citation chips and separated recommendations.
- **Problem:** Generic LLM hallucinations, fabricated policy figures, and untraceable advisory recommendations in executive decision memos.
- **Why built this way:** Hard code invariant strips non-retrieved citation IDs, separates recommendations from factual basis, and features a deterministic demo fallback ensuring 100% auditability without third-party API keys.

## 3. Sources & Evidence

**AUDIT REGISTER**




- **What it does:** Comprehensive provenance registry detailing publisher, publication date, verification status, confidence rating (high/med/low), and reverse "where cited" cross-links.
- **Problem:** Opaque market figures, zombie statistics, and conflicting definitions (e.g. 2,117 GCC parents vs 3,728 units) with zero provenance.
- **Why built this way:** Establishes defensibility and user trust before monetization; automated seed linter halts builds on sourceless numeric claims.

## 4. GCC Intelligence Dashboard

**ECOSYSTEM OVERVIEW**




- **What it does:** High-level ecosystem overview tracking 46 verified GCCs across 12 cities and 11 sectors, paired with an active data freshness ledger and policy status tracker.
- **Problem:** Conflation of sampled directory records with national market census figures, leading to misleading location business cases.
- **Why built this way:** Prominent dataset-scope banner and verification ledger distinguish tracked sample data from the national 2,117 universe.

Screenshots captured from live prototype build (Compare verified with Bengaluru 38 tracked / Karnataka policy). Module cards communicate core functional JTBD & design rationale.

GO-TO-MARKET SEQUENCING [most steps are hypotheses]

# Positioning and stakeholder adoption strategy

**Category we aim to create:** "evidence-first India GCC location intelligence" — decision layer above directories, below bespoke consulting. We never out-claim Nasscom–Zinnov/CBRE; their data is our source of record (competitive-landscape §4).

### 1. IPAs share

Free, cited compare links let state IPAs position themselves fairly vs peer states — they become our distribution catalysts (no ad spend needed, S05–S19; S12 shows IPAs actively pitch at NASSCOM events).

### 2. Providers inspect

Provider/BD leads verify market claims pre-meeting (this class already pays $3.5K–$10K for similar surfaces, S30 — organic awareness follows the free tier).

### 3. Expansion desks adopt

Priya-style desks self-serve pre-work, arrive at consultant conversations evidence-armed; conversion to paid only AFTER free-tier trust (M1/M3 instrumented).

**What we deliberately DON'T do:** no paid ranking anywhere facts are shown (trust contract), no bulk contact-data resale (O6, name/legal risk S38), no interviews implied in any deck claim (Phase-3 validates all [inference] tags).

ALL PRICING = [proposal], UNVALIDATED

# Business-model hypotheses and go-to-market

**H1 · Free core (now)**

City compare + policy matrix + source register stay free. Reputation density is the wedge; paying for data pre-trust kills it (business-model §1).

**H2 · Pro seats [proposal]**

$99–$199/yr individual; $500–$2,000/yr teams of 3–5 — anchored at 3–6% of market comparables' $3.5K–$10K cycles (S30). Gate: free-tier traction first.

**H4 · Reports [proposal]**

Free quarterly standard; $299–$499 thematic deep-dives once publishing muscle exists (market precedent: free quarterly exists S31; historic $9,999 static lists S33).

**H5 · Provider verified [proposal]**

$199–$499/yr verified-profile attributes only — zero influence on rankings/answers; banned-paid-ranking is an immutable product rule (business-model §4).

**GTM hypotheses [proposal, no spend]:** content-catalyst loop (share links) + expansion-desk outreach + token funnel metrics (M3 search→detail ≥40%; M1 median <10 min — BOTH are unvalidated targets until Phase 3 instrumentation reports).

# Data trust, AI reliability, risks and success metrics

## Trust contract (enforced in code, not just values)

- **Every figure sourced:** seed linter FAILS the build on a sourceless value (F6); provenance chips on all numeric surfaces.
- **Denominators labelled:** "GCCs, parent vs units" named at every count (S01/S02-style conflict rendered, never averaged).
- **Conflicts co-shown:** $98.4B (S02) vs $164B (S04) both visible, flagged as contested.
- **Missing ≠ zero:** "not yet tracked", draft policies flagged; placeholder sources labelled "stale".
- **AI cites or declines:** only retrieved evidence citable; unknown citation IDs stripped in tests; insufficiency is an honest low-confidence path; recommendations never blended into facts.

## Risks + metrics (targets are [proposals])

- **Coverage honesty:** 46-company tracked subset; never billed as "all 2,117 GCCs" (S02/S30 context).
- **Name collision:** gcc-atlas.com exists (S38) — working-title label everywhere; trademark check is a Phase-3 gate.
- **Data freshness:** verify-by dates shown; manual re-verification pass scheduled at submission.

| | |
|---|---|
| **M1** | Time-to-answer < 10 min [proposal] |
| **M2** | 100% of displayed figures carry sources (hard invariant); assistant ≥95% answered w/ ≥1 citation [proposal] |
| **M3** | Search→detail ≥40% [proposal] |
| **M4-M6** | compare/assistant/provider engagement + no-paid-ranking guardrail countermetrics |

PHASE GATES (product/roadmap.md)

# Roadmap, prototype and next steps

### Phase 2 — DONE: working prototype

- Research → strategy → full build (Phases 1–8)
- Retrieval→citation pipeline + demo fallback
- Rate limits, events, session ACL, QA report (108/108 tests green)
- Deployment guide + render.yaml + this deck

### Phase 3 — validation loop

- 0→1 interviews to replace [inference] tags (5+ expansion desks, 2+ IPAs)
- Re-calibrate M1/M3 targets from real funnel data
- Trademark check (S38) → rename-or-keep decision

### Phase 4 — paid options (gated)

- H2 seats only after free-traction gate; H3 workspaces; H5 provider profiles under neutrality rules
- Monitoring layer F17 if O2 survives falsification tests

**The bet, honestly framed:** the market is undeniably real and growing; the intelligence layer above it is fragmented and expensive; a discipline-first free comparator is buildable and differentiated. What we have NOT proven: demand (no interviews), willingness-to-pay (proposals only), and defensibility against a fast follower. Every one of those is instrumented to be falsifiable in Phase 3 — which is the point.

Prototype + demo script: deployment/demo-walkthrough.md · QA evidence: qa/quality-report.md · Full evidence register: research/source-register.csv · Deck sources: product/deck-evidence-sources.md